

DSC6009: MACHINE LEARNING AT SCALE

Effective Term

Semester A 2026/27

Part I Course Overview

Course Title

Machine Learning at Scale

Subject Code

DSC - Data Science

Course Number

6009

Academic Unit

Data Science (DS)

College/School

College of Computing (CC)

Course Duration

One Semester

Credit Units

3

Level

P5, P6 - Postgraduate Degree

Medium of Instruction

English

Medium of Assessment

English

Prerequisites

Nil

Precursors

Nil

Equivalent Courses

SDSC6009 Machine Learning at Scale

Exclusive Courses

Nil

Part II Course Details

Abstract

This course teaches the underlying principles required to develop scalable machine learning pipelines for structured and unstructured data at the petabyte scale. The course covers principles of scaling machine learning process under big data via

deploying the MapReduce parallel computing. In addition, the hands-on algorithmic design and development of machine learning algorithms in parallel computing environments (Spark) will be discussed. Students will use MapReduce parallel computing frameworks for machine learning in industrial applications and deployments for various fields, including advertising, finance, healthcare, and search engines.

Course Intended Learning Outcomes (CILOs)

CILOs		Weighting (if app.)	DEC-A1	DEC-A2	DEC-A3
1	Describe principles of scalable machine learning and parallel computing	20	x		
2	Discuss big data management tools and ecosystem	20	x		
3	Design and develop parallel computing and scalable machine learning algorithms	20	x	x	
4	Conduct assessment, comparison, and selection for scalable learning models	20	x	x	
5	Implement parallel compute frameworks for industrial applications	20		x	x

A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

Learning and Teaching Activities (LTAs)

LTAs		Brief Description	CILO No.	Hours/week (if applicable)
1	Lecture	Students will engage in formal lectures to gain knowledge about principles of scalable machine learning pipelines covered in this course	1, 2, 3, 4	26 hours/semester
2	Laboratory work	Students will participate in lab activities to develop the ability of implementing scalable machine learning pipelines	2, 3, 4, 5	13 hours/semester

Assessment Tasks / Activities (ATs)

	ATs	CILO No.	Weighting (%)	Remarks ("- " for nil entry)	Allow Use of GenAI?
1	Group Project	2, 3, 4, 5	40	-	Yes
2	Individual Coursework	1, 2, 3, 4	25	-	Yes

Continuous Assessment (%)

65

Examination (%)

35

Examination Duration (Hours)

2

Minimum Examination Passing Requirement (%)

30

Additional Information for ATs

For a student to pass the course, at least 30% of the maximum mark for the examination should be obtained

Assessment Rubrics (AR)

Assessment Task

Group Project (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

40%

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Individual Coursework (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

25%

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Examination (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

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(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Group Project (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

40%

Excellent

(A+, A, A-) High

Good

(B+, B) Moderate

Marginal

(B-, C+, C) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Individual Coursework (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

25%

Excellent

(A+, A, A-) High

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Assessment Task

Examination (for students admitted from Semester A 2022/23 to Summer Term 2024)

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(B-, C+, C) Basic

Failure

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Part III Other Information

Keyword Syllabus

- A review of big databases: Distributed file storage, Hadoop, Spark, MLlib
- Machine learning under big data environment: Implement machine learning methods via Spark to analyse big data, Principles in decomposing large-scale learning tasks into distributed individual sub-learning tasks, Optimization contents in distributed learning.
- Transfer learning: Domain source and target source learning; transfer learning methods, residual function transfer, discrepancies between domain source model and target source model, industrial case studies using transfer learning.
- Recommendation Systems at Scale: Graph-networks, Link Analysis, collaborative filtering, Sparsity and Scalability in recommendation systems.
- Introductory Real-time Computer Vision: Organization of training image samples, Transfer learning in CNN, You Only Look Once method.
- Programming in Spark will be covered in the lab sessions.

Reading List

Compulsory Readings

Title	
1	Lecture Notes

Additional Readings

Title	
1	Jure Leskovec, Anand Rajaraman, Jeff Ullman, Mining of Massive Datasets
2	Sandy Ryza, Uri Laserson, Sean Owen & Josh Wills. Advanced Analytics with Spark
3	Ron Bekkerman, Mikhail Bilenko, John Langford. Scaling up Machine Learning: Parallel and Distributed Approaches